

Aim:-

To determine the Voltage in the circuit Using Nodal analysis both theoretically and practically for a given DC circuit

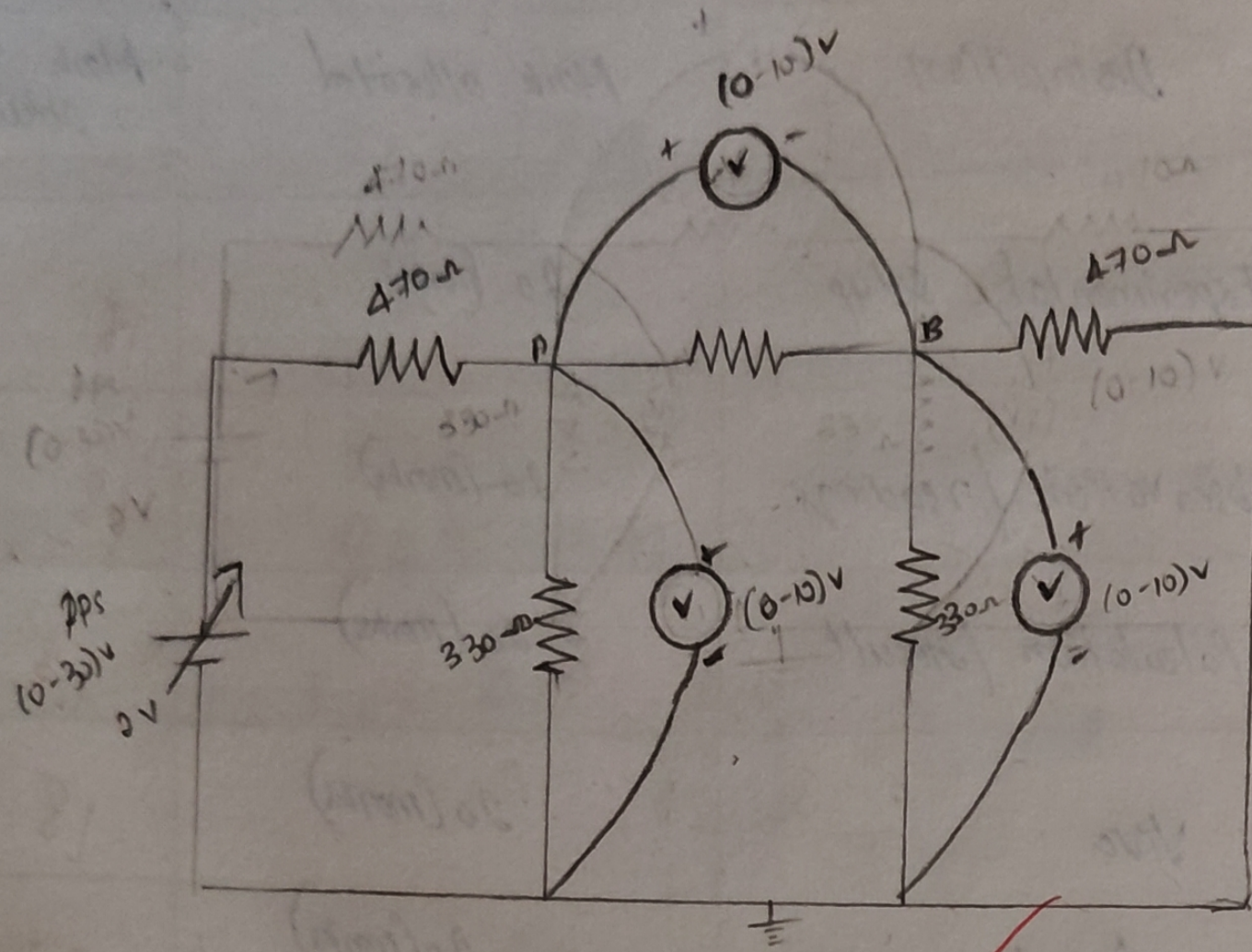
APPARATUS REQUIRED:-

S.L. No	APPARATUS	SPECIFICATION	QUANTITY
1	Regulated Power supply (RPS)	(0-30)V	1
2	Multimeter	-	1
3	Resistors	470- Ω , 330- Ω	3-2
4	Bread board	-	1

THEORY:-

In electric circuit analysis, nodal analysis, Node Voltage analysis, or the branch current method is a Method of determining the Voltage (potential difference) between "nodes" (points where elements or branches connect) in an electrical circuit in terms of the branch currents.

Circuit Diagram:



Thus, the mesh analysis is verified practically and theoretically. The theoretical current for voltage supply are:

- 1. The current across 330Ω resistor is 1.5mA
- 2. The current across 470Ω resistor is 4.5mA
- 3. The current across 470Ω resistor is 4.5mA
- 4. The current across 330Ω resistor is 1.5mA

Tabular Column:-

Parameters	Theoretical	Practical
V_1	0.63	0.66
V_2	0.18	0.19
V_3	0.45	0.47

CALCULATION:-

Nodal analysis:
Node A
By KCL:

$$\frac{V_A - 2}{470} + \frac{V_A}{330} + \frac{V_A - V_B}{470} = 0$$

$$\frac{33V_A - 66 + 47V_A + 33V_A - 33V_B}{15570} = 0$$

$$113V_A - 33V_B = 66 \rightarrow \textcircled{1}$$

At Node B

By KCL:

$$\frac{V_B - V_A}{470} + \frac{V_B}{330} + \frac{V_B}{470} = 0$$

$$33V_B - 33V_A + 47V_B + 33V_B = 0$$

PROCEDURE:-

1. Give connections as per the circuit diagram
2. Switch ON the Supply, vary the (Regulated Power supply) and set a particular input voltage.
3. Note down the readings of ammeters and voltmeters and tabulate them.
4. Vary the Aps for different input voltage and note down the readings of all the meters
5. Reduce the Aps to its minimum value and switch OFF the supply.
6. Using the tabulated values, verify Kirchhoff's laws practically, and verify it theoretically.

$$113V_B - 33V_A = 0$$

$$113V_B = 33V_A$$

$$V_A = \frac{113V_B}{33} \rightarrow (2)$$

Sub (2) in (1)

$$113 \left(\frac{113V_B}{33} \right) - 33V_B = 66$$

$$\frac{12769}{33} V_B - \frac{1089}{33} V_B = 66$$

$$V_B = \frac{2178}{11680}$$

$$V_B = 0.186V$$

$$V_A = \frac{113}{33} V_B = \frac{113}{33} \times 0.18$$

$$V_A = 0.638V$$

$$V_1 = V_A = 0.63V$$

$$V_2 = V_B = 0.18V$$

$$V_3 = V_A - V_B = 0.63 - 0.18$$

$$= 0.45V$$

Sr. No	Description	Marks allocated	Mark obtained
1	Experimental setup	20 (marks)	18
2	Observation / readings	20 (mark)	18
3	Calculation / result	20 (mark)	18
4	Viva	20 (mark)	18
5	record	20 (mark)	20

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RESULT:-

Thus the Nodal analysis is Verified practically and theoretically. The resultant Voltages for V Supply are

- The Voltage V_1 is $0.63V$
- The Voltage V_2 is $0.18V$
- The Voltage V_3 is $0.45V$